DelftX: MathMod1x Mathematical Modelling Basics

Help

Validation

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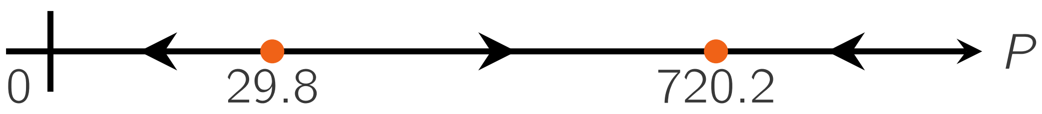
The initial value problem describing the fish population with bounded growth is:

$$\frac{dP}{dt} = 0.7\,P\left(1 - \frac{P}{750}\right) - 20, \quad P(0) = 30.$$

The phase line of this autonomous differential equation is:



Exercise A

1/1 point (graded)

For this initial value problem $\lim_{t\to\infty} P(t)$ equals

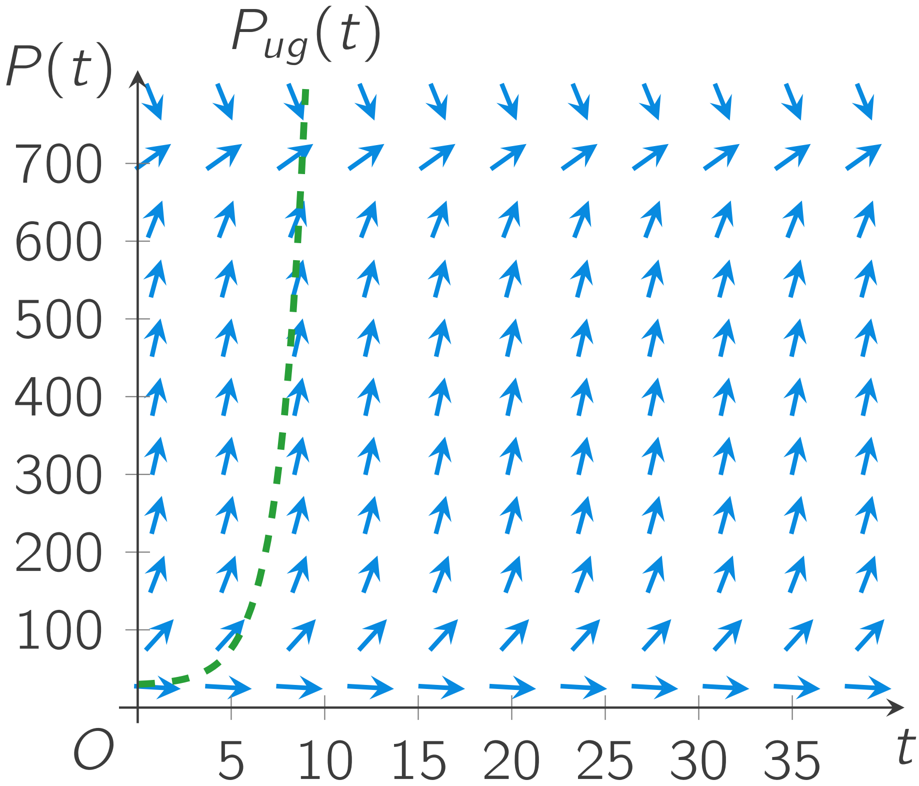
- ∞, because the population will grow for ever.
- 750, because that is the maximum capacity of the aquarium.
- 720.2, because in that equilibrium point, the increase by the new fish exactly balances the 20 fish sold every day, and the equilibrium point is stable. ✔
- 29.8, because the increase by the new fish exactly balances the 20 fish sold every day.
- 0, because more fish are sold than new fish hatch, so after some time, the population dies out.

Submit You have used 1 of 2 attempts

Answers are displayed within the problem

Here, you see the direction field for P, when

$$\frac{dP}{dt} = 0.7\,P\left(1 - \frac{P}{750}\right) - 20.$$



For comparison, the graph of $P_{ug}(t)$ is shown as well. $P_{ug}(t)$ is the solution with the unbounded growth, we used in Module 1.

$$\frac{dP_{ug}}{dt} = 0.7\,P_{ug} - 20, \quad P_{ug}(0) = 30.$$

Exercise B

1/1 point (graded)
$P_{ug}(t)$ reaches 720 after about 9 days.

Estimate from the direction field, after how many days $P(t)$ reaches 720 rainbowfish.

25 ✔ **Answer:** 30

25

Answer
Correct:
It is very difficult to guess this from a direction field. You can only try to follow the arrows from $P(0) = 30$ onwards. It should take longer than the 9 days of P_{ug}, and probably shorter than 50 days or so, but how much longer is anybodies guess.

Submit You have used 1 of 2 attempts

Answers are displayed within the problem

In Section 2.3 you will learn a method to simulate solutions of a differential equation. Those simulations predict the solution much more precisely than what you can achieve by eye-balling the direction field. But first, a page for the questions you might have.